Moment Tensor Decomposition

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1 Physical Properties of the Decomposition

Vavryčuk (2015) discusses the decomposition of a general seismic moment tensor into three elementary components: the isotropic component (ISO), the double-couple component (DC), and the compensated linear vector dipole component (CLVD). This decomposition is mainly used to classify seismic source types and to provide a physical interpretation of non-double-couple mechanisms.

The moment tensor can be written schematically as

$$\mathbf{M} = \mathbf{M}_{ISO} + \mathbf{M}_{DC} + \mathbf{M}_{CLVD}.$$

In normalized form, the relative scale factors are usually denoted as

$$C_{ISO}, \qquad C_{DC}, \qquad C_{CLVD},$$

where

$$|C_{ISO}| + |C_{CLVD}| + C_{DC} = 1.$$

Here, C_{DC} is non-negative and ranges from 0 to 1, while C_{ISO} and C_{CLVD} may be positive or negative.

1.1 Isotropic Component: ISO

The isotropic component represents a purely volumetric source. It corresponds to an equal expansion or contraction in all directions.

A positive isotropic component,

$$C_{ISO} > 0,$$

is associated with an explosive or opening-type source. Physically, this means that the source volume increases. Typical examples include explosions or tensile crack opening.

A negative isotropic component,

$$C_{ISO} < 0,$$

corresponds to an implosive or closing-type source. This means that the source volume decreases. Examples may include cavity collapse, compaction, or compressive crack closure.

A pure isotropic source is characterized by

$$C_{ISO} = \pm 1, \qquad C_{DC} = 0, \qquad C_{CLVD} = 0.$$

Therefore, in the source-type plot, pure explosions and implosions occupy the top and bottom vertices of the diamond plot, respectively.

1.2 Double-Couple Component: DC

The double-couple component is the most familiar component in earthquake seismology. It represents shear slip on a planar fault in an isotropic medium.

A pure double-couple source is characterized by

$$C_{DC} = 1, \qquad C_{ISO} = 0, \qquad C_{CLVD} = 0.$$

This corresponds to conventional tectonic earthquakes, where the source process is dominated by shear faulting without any net volume change.

In the CLVD–ISO diamond plot, a pure DC source is located at the origin:

$$C_{ISO} = 0, \qquad C_{CLVD} = 0.$$

Physically, the DC component is the part of the moment tensor that measures the amount of shear faulting. This is why Vavryčuk emphasizes that a physically meaningful decomposition should assign zero DC to purely tensile or compressive cracks, because such cracks contain no shear motion.

1.3 Compensated Linear Vector Dipole Component: CLVD

The CLVD component is a non-double-couple deviatoric component. Unlike ISO and DC, it does not correspond to a simple elementary physical source by itself. However, it is mathematically necessary to complete the decomposition of a general moment tensor.

A pure CLVD source has

$$C_{CLVD} = \pm 1, \qquad C_{ISO} = 0, \qquad C_{DC} = 0.$$

However, Vavryčuk points out that pure CLVD does not have a straightforward physical interpretation comparable to explosion, implosion, or shear faulting. Instead, CLVD becomes physically meaningful when interpreted together with the ISO component.

For example, tensile or compressive faulting generally produces both ISO and CLVD components. In this case, the CLVD component reflects the deviatoric part of the crack-type deformation, while the ISO component reflects the associated volume change.

For tensile faulting, both components are positive:

$$C_{ISO} > 0, \qquad C_{CLVD} > 0.$$

For compressive faulting, both components are negative:

$$C_{ISO} < 0, \qquad C_{CLVD} < 0.$$

Therefore, CLVD should not always be interpreted as an independent physical source. Instead, it often appears as part of a more complex physical mechanism, especially tensile or shear-tensile faulting.

1.4 Pure Tensile and Pure Compressive Faulting

A pure tensile or compressive crack is free of shear motion. Therefore, according to Vavryčuk, it should have

$$C_{DC} = 0.$$

However, it generally contains both ISO and CLVD components:

$$C_{ISO} \neq 0, \qquad C_{CLVD} \neq 0.$$

The ratio between the isotropic and CLVD components depends on the elastic properties of the medium surrounding the source. In isotropic media, Vavryčuk gives the relation

$$\frac{C_{ISO}}{C_{CLVD}} = \frac{3}{4}\left(\frac{v_P}{v_S}\right)^2 - 1.$$

This means that the position of a tensile or compressive crack in the CLVD–ISO diamond plot depends on the v_P/v_S ratio of the medium.

1.5 Shear-Tensile Sources

A shear-tensile source combines shear faulting and tensile opening or closing. Therefore, it generally contains all three components:

$$C_{ISO} \neq 0, \qquad C_{DC} \neq 0, \qquad C_{CLVD} \neq 0.$$

Positive values of C_{ISO} and C_{CLVD} correspond to tensile mechanisms, where the fault opens during rupture. Negative values correspond to compressive mechanisms, where the fault closes during rupture.

In this type of source, the relative amount of non-DC components compared with the DC component provides information about the angle between the slip vector and the fault plane. If the slip vector lies exactly within the fault plane, the source is a pure shear source. If the slip vector has a component normal to the fault, the source becomes shear-tensile or shear-compressive.

1.6 Shear Faulting on a Non-Planar Fault

Vavryčuk also notes that shear faulting on a non-planar, curved, or irregular fault may produce non-zero CLVD components. However, because no net volume change is associated with this type of source, the isotropic component is expected to be zero:

$$C_{ISO} = 0, \qquad C_{DC} \neq 0, \qquad C_{CLVD} \neq 0.$$

Thus, a non-zero CLVD component does not necessarily imply tensile opening or closing. It may also reflect geometric complexity of the faulting process.

2 Interpretation in the CLVD–ISO Diamond Plot

The decomposition can be visualized using the CLVD–ISO diamond plot. In this plot, the horizontal axis represents C_{CLVD}, and the vertical axis represents C_{ISO}. The DC component is not plotted as a separate axis, but it is implied by the distance from the center because

$$C_{DC} = 1 - |C_{ISO}| - |C_{CLVD}|.$$

The main physical interpretations are:

$$(C_{CLVD}, C_{ISO}) = (0, 0)$$

corresponds to a pure DC source, i.e., pure shear faulting.

Points along the CLVD axis correspond to sources with no isotropic component, such as shear faulting on non-planar faults.

Points in the first and third quadrants, where C_{ISO} and C_{CLVD} have the same sign. Positive ISO and CLVD indicate tensile opening, while negative ISO and CLVD indicate compressive closing.

Vavryčuk also notes that the second and fourth quadrants are not usually occupied by the basic source types discussed above. Moment tensors located in these quadrants may indicate inversion errors, noise, limited station coverage, inaccurate velocity models, complicated source processes, or anisotropy in the focal region.

3 Difference Between Moment Tensor Decomposition and Source Tensor Decomposition

A key point is that moment tensor decomposition and source tensor decomposition are not always equivalent. Their physical meaning depends strongly on whether the medium is isotropic or anisotropic.

3.1 Moment Tensor Decomposition

The moment tensor $\mathbf{M}$ describes the equivalent body forces radiated by a seismic point source. It is the quantity commonly obtained from seismic waveform inversion.

In isotropic media, the moment tensor can often be physically interpreted in terms of faulting geometry. For example, a pure shear fault on a planar fault produces a pure DC moment tensor. A tensile crack produces ISO and CLVD components. Therefore, in isotropic media, moment tensor decomposition provides a useful classification of the source mechanism.

However, in anisotropic media, the moment tensor depends not only on the source geometry but also on the elastic properties of the medium. The relation between the source process and the observed moment tensor becomes more complicated. Even simple shear faulting on a planar fault can produce a moment tensor with non-zero ISO, DC, and CLVD components.

This means that in anisotropic media, non-DC components in the moment tensor do not necessarily imply a non-shear source. They may instead be caused by elastic anisotropy near the source.

3.2 Source Tensor Decomposition

The source tensor, also called the potency tensor, is more directly related to the geometry of faulting. Vavryčuk defines the source tensor for a dislocation source as

$$D_{kl} = \frac{uS}{2}\left(s_k n_l + s_l n_k\right),$$

where u is the slip, S is the fault area, $\mathbf{n}$ is the fault normal, and $\mathbf{s}$ is the slip direction.

Unlike the moment tensor, the source tensor describes the actual deformation at the source. Therefore, it is directly connected to the fault geometry and the slip direction.

In anisotropic media, the moment tensor and source tensor are related by the elastic stiffness tensor:

$$M_{ij} = c_{ijkl} D_{kl}.$$

In isotropic media, this relation simplifies to

$$M_{ij} = \lambda D_{kk} \delta_{ij} + 2\mu D_{ij},$$

where λ and μ are the Lamé parameters.

3.3 Main Conceptual Difference

The main difference is that the moment tensor includes the effect of both source geometry and medium properties, whereas the source tensor describes the source geometry more directly.

This distinction can be summarized as follows:

$$\text{Moment tensor} = \text{source process} + \text{elastic properties of the medium.}$$

By contrast,

$$\text{Source tensor} = \text{geometry of faulting and slip.}$$

In isotropic media, the eigenvectors of the moment tensor and source tensor are the same, and their decompositions lead to similar physical interpretations. Therefore, moment tensor decomposition is usually sufficient.

In anisotropic media, however, the moment tensor and source tensor may have different eigenvectors and different decompositions. As a result, moment tensor decomposition alone may be misleading. A non-zero ISO or CLVD component in the moment tensor may reflect anisotropic elastic properties rather than true tensile opening, closing, or source complexity.

Therefore, Vavryčuk argues that in anisotropic focal regions, physical interpretation should be based on source tensor decomposition rather than moment tensor decomposition.

4 Summary

The ISO–DC–CLVD decomposition is a useful tool for classifying seismic sources. The ISO component represents volumetric change, with positive values corresponding to explosion or opening and negative values corresponding to implosion or closing. The DC component represents shear faulting on a planar fault and is the dominant component of ordinary tectonic earthquakes. The CLVD component is mathematically required for a complete decomposition of a general moment tensor, but it does not usually represent a simple independent physical source. Instead, it is often physically meaningful when combined with the ISO component, especially in tensile, compressive, or shear-tensile faulting.

In isotropic media, moment tensor decomposition can be interpreted directly in terms of physical source mechanisms. Pure shear faulting gives a dominant DC component, explosions and implosions give ISO components, and tensile or compressive cracks generate coupled ISO and CLVD components. However, in anisotropic media, the moment tensor is affected by the elastic properties of the medium. Consequently, even simple faulting can generate apparent non-DC components. For this reason, Vavryčuk emphasizes the distinction between moment tensor decomposition and source tensor decomposition. The source tensor is more directly related to fault geometry and slip, while the moment tensor is the elastic response of the medium to that source. Therefore, for anisotropic focal regions, source tensor decomposition provides a more physically reliable interpretation of the earthquake source than moment tensor decomposition alone.